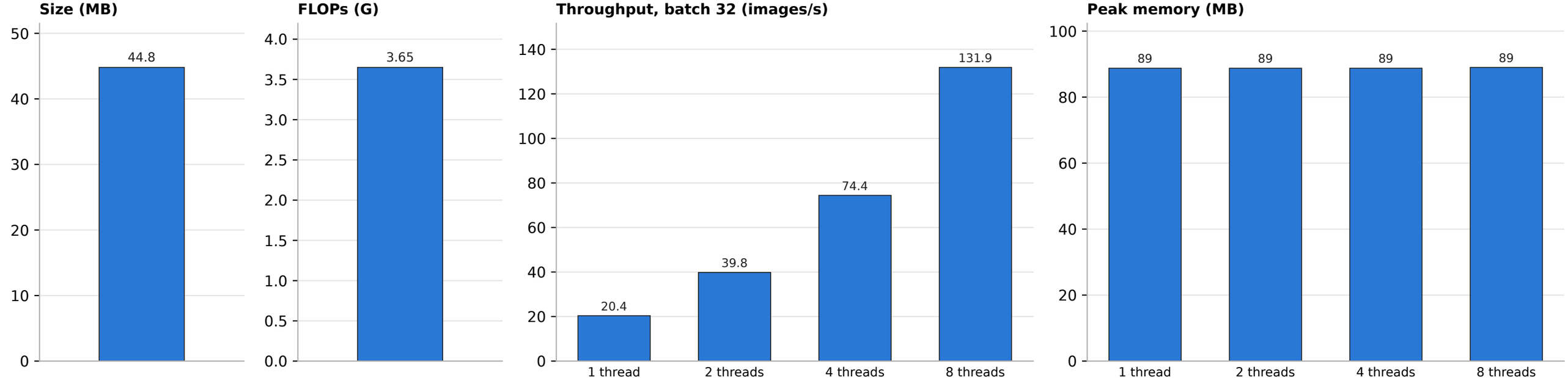FP32: efficiency (validation)